ARXIS

AEC Architecture Traceability & Engineering Change Governance

Graph-Theoretic Dependency Management for Complex Built-Asset Infrastructure

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Live Case Study | GitHub Repository
Methodological Alignment: ISO 19650 Principles | FIDIC Change Governance

Executive Summary & Strategic Context

In large-scale Architecture, Engineering, and Construction (AEC) capital projects, uncoordinated design revisions represent the single greatest cause of budget overruns and contractor delay claims. On-site physical rework accounts for 5% to 12% of total project expenditure, frequently triggered when a localized architectural change invalidates structural load limits or MEP (Mechanical, Electrical, Plumbing) routings without cross-disciplinary synchronization.

ARXIS is an open-source design-traceability reference engine authored by **Igor Grybek** to evaluate dependency governance across capital project documentation. Grounded in infrastructure capex experience and CIB audit discipline, ARXIS operates under the institutional axiom: *traceability does not grant approval*. Operating as a compact Python reference prototype (47 LOC), ARXIS maps requirements, engineering decisions, and documentation artifacts into a tabular specification linkage, auditing version drift and evaluating downstream change impact prior to executive sign-off.

System Architecture & Graph Mechanics

The platform coordinates capital project information flows across four mathematical tiers:

1. **Tier 1: Client Requirements:** Formalizes functional, environmental (LEED), and regulatory mandates into unique nodes (REQ_ID). Establishes unambiguous root intent.
2. **Tier 2: Engineering Decisions:** Maps trade-off decisions made by lead engineers across disciplines (DEC_ID). Links choices directly to governing requirement IDs and registered approval authorities.
3. **Tier 3: Design Deliverables (Artifacts):** Models design documentation and technical schedules (ART_ID) designed for downstream IFC/BIM schema alignment. Binds each deliverable to its upstream decision and SHA-256 content hash. Strictly eliminates orphan deliverables.
4. **Tier 4: Topological Change Impact Engine:** When an Engineering Change Request (ECR) is evaluated, the reference engine traverses dependency mappings to compute the downstream impacted set $\mathcal{R}(u)$. All dependent documents are flagged as `PENDING_REEVALUATION`, modeling audit holds before site execution.
5. **Tier 5: RACI Authorization Gates:** Enforces role-based sign-offs. Rejects baseline releases unless certified discipline leads (Architectural, Structural, MEP) have approved all linked decisions.

5-Dimensional Multidimensional Operating Model

In accordance with the 5-Dimensional Multidimensional Operating Model, ARXIS integrates five operational dimensions:

- **Technical Architecture:** Python standard library specification linkage (dictionary and set mapping), deterministic topological dependency traversal, OpenAPI 3.1 REST contracts, and reproducible execution manifests.
- **Operating Workflow (RACI):** Project Director (authorizes change budgets) $\rightarrow$ Discipline Leads (verify technical feasibility) $\rightarrow$ Document Controller (locks Common Data Environment baselines) $\rightarrow$ General Contractor (verifies field installation against locked version hashes).
- **Financial Model & Governance Economics:** Evaluates structured change coordination effort (€250–€490 per change event) against benchmark physical clash rework expenses (€50,000–€250,000 per site clash), modeling the substantial downside mitigation value of pre-execution documentation reviews.
- **Risk & Governance Model:** Methodological alignment with ISO 19650 (BIM information containers and state transitions) and international construction change principles (FIDIC Red/Yellow Books, AIA Document G701).
- **Scaling & Integration:** Conceptual architecture designed for open BIM schemas (IFC 4.3) and Common Data Environment (CDE) webhook synchronization.

Benchmark Performance (Synthetic 3-Tier Dependency Fixture)

The reference implementation is verified via 7 automated regression test suites on synthetic dependency fixtures in the public repository (arxis-traceability on GitHub):

Benchmark Metric	Observed Score	Traditional Baseline	Operational Significance
Graph Dependency Coverage	**100.0%**	60.0–75.0%	Deterministic verification: zero orphan decisions or deliverables.
Cycle Detection Latency	$<$ **5 ms**	Manual / Days	Deterministic traversal instantly detects and halts circular dependencies.
Blast Radius Traversal	**100.0%**	Subjective	Transitive closure tags every downstream affected drawing.
Unauthorized Actor Blocks	**100.0%**	0.0% (Informal)	Prevents unauthorized verbal approvals; logs security faults.
Illustrative Clash ROI	**Modeled**	1.0×	Literature heuristic: early review mitigates exponential site rework.

Architecture Decision Records (ADRs) & API Contracts

All architectural trade-offs are formally governed by RFC-style Architecture Decision Records:

- **ADR-0001:** Directed Acyclic Graph (DAG) for Change Blast Radius Traversal.
- **ADR-0002:** ISO 19650 Common Data Environment (CDE) Deliverable Locking & Hash Manifests.
- **ADR-0003:** Programmatic RACI Gate Enforcement Prior to Construction Baseline Release.
- **OpenAPI 3.1 Contract:** Endpoints for ‘/v1/change-requests/submit‘, ‘/v1/impact/blast-radius‘, and ‘/v1/gates/sign-off‘.

Methodological & Confidentiality Notice: This architectural whitepaper utilizes validated synthetic benchmark datasets and open institutional AEC standards (ISO 19650, IFC 4.3, FIDIC). No confidential corporate banking or private client data from active or past institutional engagements (BNP Paribas CIB, Santander CIB) is contained herein. © 2026 Igor Grybek. All rights reserved.